

Six-degree-of-freedom cascaded fractional-order control of an over-actuated ROV: margin-constrained tuning and a bias-controlled comparison of metaheuristics

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Abstract

[TODO: Write after the results are fixed. Structure: (i) single-axis FO control of ROVs and the gap (coupled six-DOF motion, realistic thrusters, latency); (ii) the FOPID-(1+TFOID) cascade from load-frequency control, recast as a position–velocity cascade for all six DOF of a BlueROV2 Heavy with eight T200 thrusters fitted to manufacturer data (thrust and electrical power); (iii) multi-scenario tuning with a phase-margin constraint on every linearized DOF loop; (iv) the newest tuner (SOO) and five others under an equal evaluation budget, a mirrored search box and shifted benchmarks, which expose an origin bias of SOO, GJO and GWO; (v) headline numbers from T1–T4, robustness and Monte Carlo.]

Keywords: remotely operated vehicle, fractional-order control, cascade control, FOPID, tilt-integral-derivative, metaheuristic tuning, phase margin, BlueROV2

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1. Introduction

[TODO: Paragraph 1: ROV/AUV motion control, coupled nonlinear hydrodynamics, wave and current disturbances; most FO studies regulate a single axis.]

[TODO: Paragraph 2: our earlier study [1]: surge-axis $PI^\lambda D^\mu$ [2], fourth-order plant identified from T200 measurements, PSO [3] and DE [4] over a multi-scenario cost, four tests, FPGA realization. Limits: one axis, no coupling, no allocation, no latency or margin requirement.]

[TODO: Paragraph 3: cascaded FO controllers in load-frequency control; FOPID-(1+TFOID) of [5] with SOO [6]; its own stated limits (no formal stability analysis, simulation only).]

[TODO: Paragraph 4: tuning by new metaheuristics; equal-iteration comparisons that give some methods twice the evaluations; origin/centre bias [7, 8]; inconsistencies between the SOO paper and its code [9].]

The contributions of this paper are:

1. The FOPID-(1+TFOID) cascade is recast as a position–velocity cascade (the second stage acts on $u_1 - \nu$, the mechanical counterpart of $u_1 - \Delta f$) and applied to all six DOF of an over-actuated BlueROV2 Heavy, with six benchmark structures realized in the same cascade form.
2. Tuning minimizes a multi-scenario cost (six-DOF ITAE and electrical energy) subject to a phase margin of at least 45° on every linearized DOF loop, including the command latency and the thruster lag. Without this constraint, ITAE tuning returns loops with margins below 10° that one or two extra samples of latency destabilize (Sect. 6.5).

3. The thruster model (thrust, dead band, asymmetry, saturation and electrical power at 10–20 V) is fitted to the manufacturer’s measurements [10], so the effort term is energy in joules and battery voltage is a physically meaningful uncertainty.
4. Six tuners, including the newest (SOO), are compared at an equal evaluation budget, in the original and in a mirrored search box, and on shifted benchmark functions; this separates search ability from attraction to the coordinate origin.
5. The four tests of [1] are extended to six DOF and complemented by plant, actuator, latency and thruster-loss variations and a Monte Carlo study.

2. Vehicle and actuator model

2.1. Rigid-body and hydrodynamic model

Let $\boldsymbol{\eta} = [x, y, z, \phi, \theta, \psi]^\top$ be the position and attitude in the north-east-down frame and $\boldsymbol{\nu} = [u, v, w, p, q, r]^\top$ the velocities in the body frame, whose origin is the centre of gravity. The vehicle obeys [11]

$$\begin{aligned} \mathbf{M}\dot{\boldsymbol{\nu}} + \mathbf{C}_{RB}(\boldsymbol{\nu})\boldsymbol{\nu} + \mathbf{C}_A(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{D}(\boldsymbol{\nu}_r)\boldsymbol{\nu}_r + \mathbf{g}(\boldsymbol{\eta}) &= \mathbf{T}\mathbf{f} + \boldsymbol{\tau}_d, \\ \dot{\boldsymbol{\eta}} &= \mathbf{J}(\boldsymbol{\eta})\boldsymbol{\nu}, \end{aligned} \tag{1}$$

where $\mathbf{M} = \mathbf{M}_{RB} + \mathbf{M}_A$ is diagonal, $\boldsymbol{\nu}_r = \boldsymbol{\nu} - [\mathbf{R}^\top \mathbf{v}_c; \mathbf{0}]$ is the velocity relative to an irrotational current $\mathbf{v}_c$ (the term $\mathbf{M}_A \dot{\mathbf{v}}_c$ is neglected), $\mathbf{D}(\boldsymbol{\nu}_r) = \mathbf{D}_l + \mathbf{D}_q \text{diag}(|\boldsymbol{\nu}_r|)$, and $\mathbf{g}$ contains the restoring forces of the weight and buoyancy. The parameters of the heavy configuration are those of the open benchmark of [12] (Table 1).

Table 1: BlueROV2 Heavy parameters [12].

mass m	13.5 kg	volume ∇	0.0135 m ³ (neutral)
I_x, I_y, I_z	0.26, 0.23, 0.37 kg m ²	$\mathbf{r}_b$	[0, 0, -0.01] m
added mass	$X_{\dot{u}}$ 6.357, $Y_{\dot{v}}$ 7.121, $Z_{\dot{w}}$ 18.69 kg; $K_{\dot{p}}$ 0.1858, $M_{\dot{q}}$ 0.1348, $N_{\dot{r}}$ 0.2215 kg m ²		
linear damping	X_u 13.7, Y_v 0, Z_w 33 N s/m; K_p 0, M_q 0.8, N_r 0 N m s/rad		
quadratic damping	$X_{u u }$ 141, $Y_{v v }$ 217, $Z_{w w }$ 190 N s ² /m ² ; $K_{p p }$ 1.192, $M_{q q }$ 0.47, $N_{r r }$ 1.5 N m s ² /rad ²		

2.2. Thrusters and allocation

Four horizontal thrusters at 45° and four vertical thrusters give the 6×8 allocation matrix $\mathbf{T}$ of [12]. A demanded generalized force $\boldsymbol{\tau}$ is distributed by $\mathbf{f}_d = \mathbf{T}^+ \boldsymbol{\tau}$ and scaled down as a whole when a thruster limit is reached, so the direction of $\boldsymbol{\tau}$ is kept. Each command is mapped to a normalized PWM value $u_i \in [-1, 1]$ by the inverse of the nominal 16 V curve, with dead-band compensation. The thrust produced is

$$F(u) = \begin{cases} \sum_{k=1}^3 a_k (u - d_+)^k, & u > d_+, \\ 0, & -d_- \leq u \leq d_+, \\ -\sum_{k=1}^3 b_k (-u - d_-)^k, & u < -d_-, \end{cases} \quad (2)$$

followed by a first-order lag τ_m [TODO: replace the assumed $\tau_m = 0.1$ s by the T200 dynamics identified in [1]]. The coefficients were fitted by least squares to the manufacturer’s measurements at 10–20 V [10]; at 16 V the fit error is 0.24 N RMS (1.3 N maximum) over the range -39.9 N to 51.4 N. The electrical power of the actual thrust, $P(F) = \sum_k q_k |F|^k$, is fitted to the same data (3.1 W RMS of 389 W). The command reaches the thrusters one control period ($\Delta t = 10$ ms) after the measurement it was computed from.

2.3. Disturbances and navigation errors

Waves act as a sum of five sinusoids per DOF with random frequencies in 0.4 rad/s to 1.6 rad/s (amplitudes 20, 20, 15 N and 1 N m), and a current of 0.25 m/s ($\pm 20\%$ slow variation) acts through $\boldsymbol{\nu}_r$. The controller sees the output of a navigation filter, modelled as first-order Gauss–Markov errors with 2 Hz bandwidth (one sigma: position 1 cm, depth 5 mm, attitude 0.17–0.29°, velocity 1 cm/s, rates 0.29°/s).

3. Controller

3.1. Position–velocity form of the FOPID-(1+TFOID)

For each DOF i , the first stage acts on the position error $e_{1,i}$ (expressed in the body frame for the translational DOFs) and returns a velocity reference,

$$u_{1,i}(s) = \left(K_p + \frac{K_{i1}}{s^{\lambda_1}} + K_{d1}s^{\mu_1} \right) e_{1,i}(s). \quad (3)$$

The second stage acts on $e_{2,i} = u_{1,i} - \nu_i$ and returns an acceleration demand,

$$a_i(s) = \left(1 + \frac{K_t}{s^{1/n}} + \frac{K_{i2}}{s^{\lambda_2}} + K_{d2}s^{\mu_2} \right) e_{2,i}(s), \quad \tau_i = M_{ii} a_i. \quad (4)$$

In [5] the second stage acts on $u_1 - \Delta f$; here the local feedback is the measured velocity, which makes the cascade a position–velocity cascade in the usual sense of marine control. Scaling by the nominal inertia M_{ii} makes the gains dimensionally consistent across DOFs. One parameter set is shared by (x, y, z) and one by (ϕ, θ, ψ) : 22 decision variables.

3.2. Benchmark structures

Table 2 lists the six benchmarks. FOPID is the single-loop $\text{PI}^\lambda \text{D}^\mu$ of [1] extended to six DOF, P-PID the integer cascade of common ROV autopilots,

Table 2: Controller structures. G_1 acts on the position error; G_2 on $u_1 - \nu$ (cascades only).

Decision variables: two groups (translation, rotation).

structure	$G_1(s)$	$G_2(s)$	variables
FOPID-(1+TFOD) (proposed)	$K_p + K_{i1}s^{-\lambda_1} + K_{d1}s^{\mu_1}$	$1 + K_t s^{-1/n} + K_{i2}s^{-\lambda_2} + K_{d2}s^{\mu_2}$	22
FOPID [1]	$K_p + K_i s^{-\lambda} + K_d s^{\mu}$	–	10
PID	$K_p + K_i/s + K_d s/(s/\omega_h + 1)$	–	6
P-PID	K_{p1}	$K_{p2} + K_{i2}/s + K_{d2}s/(s/\omega_h + 1)$	8
PI-(1+FOPID)	$K_{p1} + K_{i1}/s$	$1 + K_{p2} + K_{i2}s^{-\lambda} + K_d s^{\mu}$	14
FOPID-FOPi	$K_{p1} + K_{i1}s^{-\lambda_1} + K_d s^{\mu}$	$K_{p2} + K_{i2}s^{-\lambda_2}$	16
FOPi-FOPD	$K_{p1} + K_i s^{-\lambda}$	$K_{p2} + K_d s^{\mu}$	12

and the three fractional cascades are those compared in [5]. All cascades use the position–velocity form of Eqs. (3)–(4), so the comparison isolates the stage transfer functions.

3.3. Realization

Each s^α is approximated by the Oustaloup filter [13] of order $N = 5$ on 10^{-3} rad/s to 10^2 rad/s; for $\alpha < -1$ an exact integrator is split off. Every first-order section is discretized by the Tustin rule at $\Delta t = 10$ ms. Integer derivatives use the same high-frequency limit, $K_d s/(s/\omega_h + 1)$ with $\omega_h = 100$ rad/s. The reference passes a critically damped second-order model with $\omega = 1$ rad/s, identical for all structures.

4. Tuning problem

4.1. Cost

Two training cases are used, whose disturbance and noise realizations differ from those of the tests: a simultaneous step in x , y , z and ψ (20 s),

and station keeping in waves and current with steps in ψ and z (30s). The cost is

$$J = \sum_s \left[\sum_{i=1}^6 w_i \int_0^{T_s} t |e_i(t)| dt + \rho E_s \right] + K_{pm} \sum_{i=1}^6 \frac{\max(0, \text{PM}^* - \text{PM}_i)}{\text{PM}^*}, \quad (5)$$

with $w = [1, 1, 1, 2, 2, 2]$ (one metre of error weighs as much as 0.5rad), the electrical energy E_s in joules, $\rho = 2 \times 10^{-4} \text{ J}^{-1}$, $\text{PM}^* = 45^\circ$ and $K_{pm} = 200$. Diverging runs receive $10^4(1 + \text{lost fraction})$.

4.2. Margin constraint

About hover, DOF i is $\ddot{y} + c_{1,i}\dot{y} + c_{0,i}y = H(s)a$ with $c_1 = D_{l,ii}/M_{ii}$, the restoring stiffness $c_0 = z_b B/M_{ii}$ in roll and pitch, and $H(s) = e^{-1.5\Delta t s}/(\tau_m s + 1)$ (one period of latency and the zero-order hold). Broken at a , the loop is

$$L_i(s) = G_2(s)(G_1(s) + s)P_i(s), \quad P_i(s) = \frac{H(s)}{s^2 + c_{1,i}s + c_{0,i}}, \quad (6)$$

($L_i = G_1 P_i$ for single-loop structures), with G_1, G_2 the stage transfer functions as realized (Oustaloup). PM_i is the smallest distance of $\arg L_i$ from -180° over all gain crossovers below 300rad/s; a loop whose crossover lies above (the Nyquist frequency is 314rad/s) gets $\text{PM}_i = -180^\circ$. The linear analysis predicts the latency at which the simulated loop loses stability (unit test `test_margins_predict_simulated_delay_limit`).

4.3. Search space

Gains are searched as $\log_{10} K \in [-2, \log_{10} 50]$, the orders as $\lambda \in [0, 1.5]$, $\mu \in [0, 1]$, and $n \in [1, 10]$. On a linear gain scale most random initial points violate the margin constraint and all tuners stalled near $J \approx 100$.

5. Tuners and comparison protocol

5.1. Stellar oscillation optimizer

SOO [6] moves a population of N oscillators in two stages per iteration $t = 1, \dots, T$. With the angular frequency $\omega(t) = 2\pi/(P_0 + \Delta P t)$, the scaling factor $S(t) = 2(1 - t/T)$, the best position $\mathbf{x}_b$ and $r_1, r_2, r_3 \sim U(0, 1)$ drawn per component, the first stage forms

$$\mathbf{x}_{o1} = \mathbf{x}_b - r_1 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \sin(r_2) |r_3 \mathbf{x}_b|), \quad (7)$$

$$\mathbf{x}_{o2} = \mathbf{x}_b - r_2 r_3 S(\omega r_1 - 1) (\mathbf{x}_i - |r_1 \cos(r_2) |r_3 \mathbf{x}_b|), \quad (8)$$

$$\mathbf{x}_{new} = r_3 (\mathbf{x}_{o1} + \mathbf{x}_{o2})/2, \quad (9)$$

and the second stage moves each oscillator about the mean $\bar{\mathbf{x}}_3$ of the three best,

$$\mathbf{x}_{osc} = \bar{\mathbf{x}}_3 + \frac{1}{2} [\sin(r\pi)(\mathbf{x}_a - \mathbf{x}_{b'}) + \cos((1-r)\pi)(\mathbf{x}_a - \mathbf{x}_c)], \quad (10)$$

with three other oscillators a, b', c , keeping each component of the old position with probability $1/2$. We follow the equations of the paper as read by the reimplementations in [9]: both stages in every iteration with greedy survivor selection, $P_0 = 3$, $\Delta P = 0.001$, clipping at the bounds. That source documents several differences between the paper and the authors' released code. SOO therefore spends $2N$ evaluations per iteration.

Eqs. (7)–(9) refer to the coordinate origin twice: through $|r_3 \mathbf{x}_b|$ and through the factor $r_3 \in [0, 1]$, which contracts every candidate towards $\mathbf{0}$. As a control we use *BC-SOO*, the same first stage written in coordinates

relative to $\mathbf{x}_b$ (the substitution $\mathbf{x} \mapsto \mathbf{x} - \mathbf{x}_b$),

$$\begin{aligned}\mathbf{x}_{ok} &= \mathbf{x}_b - r_k r_3 S(\omega r_1 - 1)(\mathbf{x}_i - \mathbf{x}_b), \quad k = 1, 2, \\ \mathbf{x}_{new} &= \mathbf{x}_b + r_3((\mathbf{x}_{o1} + \mathbf{x}_{o2})/2 - \mathbf{x}_b),\end{aligned}\tag{11}$$

which is translation invariant by construction.

5.2. Other tuners

PSO with the constriction coefficients of [14] ($w = 0.7298$, $c_1 = c_2 = 1.49618$, velocity limited to 20% of the range), DE/rand/1/bin [4] ($F = 0.5$, $CR = 0.9$), GWO [15], GJO [16] and uniform random search (RS). PSO and DE are the tuners of [1]; GWO and GJO are among the comparison methods of [5].

5.3. Protocol

- *Equal budget.* Every run stops after 3030 evaluations (population 30): 100 iterations for the one-stage methods and 50 for SOO. The equal-iteration protocol of [5], which gives SOO 6030 evaluations, is run as a separate setting (SOO-iter).
- *Mirrored box.* Every tuner is also run on the problem expressed in the coordinates $\mathbf{x}' = \mathbf{lb} + \mathbf{ub} - \mathbf{x}$. The problem is the same; only the position of the coordinate origin relative to the good region changes. A tuner without positional bias gives the same distribution of results in both boxes.
- *Shifted benchmarks.* Sphere, Rastrigin, Rosenbrock and Ackley functions in $D = 22$ on $[-100, 100]^{22}$, with the optimum at the origin and at a random point of $[-60, 60]^{22}$ (15 runs).

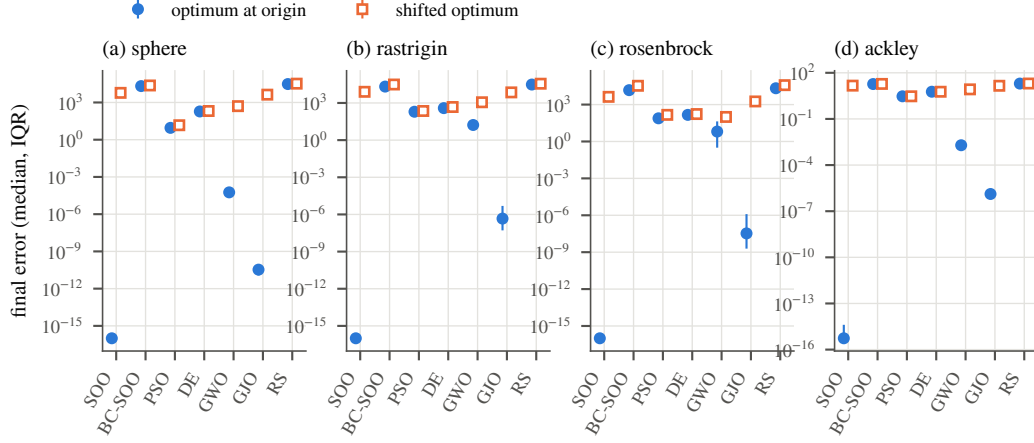


Figure 1: Final error after 3030 evaluations ($D = 22$, box $[-100, 100]^{22}$, 15 runs; median and interquartile range) with the optimum at the origin (filled) and at a random point in $[-60, 60]^{22}$ (open). Values below 10^{-15} are drawn at 10^{-16} .

- Ten independent runs per setting; medians, interquartile ranges and two-sided Mann–Whitney tests.

6. Results

6.1. Positional bias on shifted benchmarks

[TODO: Fig. 1: SOO reaches 10^{-49} on the sphere and 0 on Rastrigin and Rosenbrock with the optimum at the origin, but 6×10^3 , 8×10^3 , 4×10^3 when it is shifted; GJO and GWO behave alike; PSO and DE are unaffected. BC-SOO is translation invariant but no better than random search.]

6.2. Tuners on the six-DOF problem

[TODO: X1: table and Fig. fig_x1_runs / fig_x1_convergence.]

6.3. *Controller structures*

[TODO: X2/X3: training cost statistics; T1–T4 weighted ITAE, energy, overshoot, settling; Fig. fig_x3_tests, fig_t4_traces.]

6.4. *Robustness*

[TODO: X3: Fig. fig_robustness, Monte Carlo.]

6.5. *Margins and latency*

[TODO: X4 and X5.]

7. Discussion and limitations

[TODO: Simulation only (HIL/FPGA or pool tests next); assumed τ_m ; shared parameters per group; linearized margins about hover; navigation errors modelled, not estimated.]

8. Conclusions

[TODO:]

Data and code availability

Code, results and figure scripts: [TODO: repository URL]. The T200 data are fetched from the manufacturer and checked by SHA-256 [10].

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